Ravindra Chowdary Jonnagadla

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Education

SRM Institute of Science and Technology — Chennai, India
B.Tech in Computer Science and Engineering – Cloud Computing (CGPA: 8.01/10) — Graduated: May 2026

Technical Skills

- **Languages:** Python, C++, SQL (PostgreSQL, MySQL), JavaScript, TypeScript, Bash/Shell
- **AI & Machine Learning:** PyTorch, TensorFlow, Scikit-learn, Hugging Face, Whisper, WGAN-GP, PPO (RL), NLP
- **Cloud & DevOps:** AWS (EC2, S3, Lambda, IAM), Docker, Kubernetes, Terraform, CI/CD (GitHub Actions), Linux
- **Backend & Web:** FastAPI, Flask, Django, Node.js, Next.js, React, RESTful APIs, Microservices, Tailwind CSS
- **Specialized Domains:** Voice AI Telephony (STT/LLM/TTS), EEG Signal Processing, Hybrid Quantum-Classical Computing

Startup & Product Experience

Waani – AI Voice Calling Agent — Chennai, India
Founder & Lead Engineer — July 2026 – Present

- Architected and coded an AI voice calling agent prototype integrating telephony APIs, Sarvam AI speech models (STT/TTS), and LLM inference to evaluate natural Hindi-English code-switching dialogs.
- Built conversational state logic and dialogue branching for automated lead qualification and appointment booking, configuring fallback rules for human operator handoff.
- Engineered backend service endpoints connecting real-time audio streams with speech recognition and generation APIs, profiling latency and cost trade-offs across model providers.

LynkZen – Instagram DM Automation SaaS — Chennai, India
Founder & Lead Engineer — April 2026 – June 2026

- Built and launched an Instagram DM automation SaaS using Next.js, TypeScript, and Tailwind CSS, supporting automated comment-to-DM triggers, story replies, and keyword sequences.
- Developed an interactive visual workflow builder allowing creators to configure multi-step conditional messaging funnels and custom automated response paths without code.
- Integrated Meta Graph API messaging endpoints and grew the early-access waitlist to 3,000 signups in 2 months, against a 5,000-user capacity cap.

Selected Technical Projects

AMERS: Multimodal Emotion Recognition System | PyTorch, Signal Processing — IEEE ICETEA 2026

- Developed a multimodal emotion recognition pipeline combining raw EEG signals and speech features, implementing Cross-Modal Mutual Attention (CMMA) to capture inter-modal correlations.
- Integrated Emotion-Aware Gating (EAG) with WGAN-GP for synthetic EEG feature augmentation, applying Proximal Policy Optimization (PPO) to dynamically calibrate synthetic-to-real sampling ratios.
- Evaluated classification performance across 4 emotion classes on standard benchmarks, achieving 82.55% accuracy, 0.83 macro F1-score, and 0.77 Cohen's kappa.

SmartCap: Real-Time Subtitle Generation | Python, Whisper, FFmpeg — IEEE ICICV 2025

- Built an automated audio-visual transcription tool using OpenAI Whisper, implementing word-level acoustic timestamp alignment to generate synchronized video subtitles (.srt/.vtt).
- Engineered media preprocessing scripts using FFmpeg and Python multithreading to isolate acoustic tracks, normalize dynamic audio levels, and burn formatted subtitles onto video.

Smart QR-Based University Parking System | Flask, Python, Twilio API — IEEE CONIT 2025

- Developed a campus parking management application with dynamic, time-expiring cryptographic QR code generation for gate validation and Twilio OTP user authentication.
- Implemented a lightweight Progressive Web App (PWA) with real-time slot occupancy tracking, achieving a 98.7% booking transaction success rate across evaluated trial simulations.

Publications & Certifications

- IEEE ICETEA 2026: "Adaptive GAN-Augmented EEG–Speech Emotion Recognition via Deep Reinforcement Learning" – Int. Conf. on Emerging Technologies in Engineering Applications, Co-Author (R. C. Jonnagadla, V. Niketh, R. Naresh).
- IEEE ICICV 2025: "SmartCap: Real-Time Synchronized Subtitles for Videos" – Published in IEEE Xplore (Doc ID: 11085491), Sole/Lead Author.
- IEEE CONIT 2025: "Smart QR-Based University Parking Management System..." – Published in IEEE Xplore (Doc ID: 11166870), Co-Author.
- AWS Certified Cloud Practitioner: Amazon Web Services – Credly Verified Credential (Badge: 3dc45737).